
Putting Numbers on Chance

Random Variables, Expectation, Variance, and Joint Distributions

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So far “outcomes” have been things like heads, sunny, or spam. To do real quantitative work we attach *numbers* to outcomes — and a **random variable** is exactly that bridge. This companion shows how a random variable spreads its probability (its *distribution*), where its “centre of mass” sits (*expected value*), how spread out it is (*variance*), and what changes when two random quantities live together (*joint, marginal and conditional* distributions).

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1 Random variables: a number for every outcome

Probability so far has dealt with events like “heads” or “sunny.” But to compute averages and spreads we need numbers. A **random variable** is simply a rule that pins a number onto every outcome of an experiment.

What, why, where — for this whole session

What: a way to attach *numbers* to random outcomes, and to describe how those numbers spread their probability (a *distribution*). **Why:** numbers let us do arithmetic on chance — averages (expectation), spread (variance), and comparisons — which words like “heads” never could. **Where:** every named distribution (uniform, binomial, Poisson, Gaussian) is a random variable, and these are the raw material for the rest of the course: hypothesis testing, prediction and forecasting.

Definition 1.1: Random variable

A **random variable** X is a function from the sample space to the real numbers: it assigns exactly one number to each outcome.

- **Discrete:** takes *countable*, separated values — number of heads, number of cars in line, defects per hour.
- **Continuous:** takes *any* value in an interval — height, waiting time, temperature.

A scoreboard for chance

Think of X as a scoreboard. The experiment happens (dice roll, customer arrives), and the scoreboard instantly shows a number. “Discrete” is a scoreboard that clicks between whole values; “continuous” is a smooth dial that can rest anywhere. The quick test: *can you count the possibilities (discrete) or only measure them (continuous)?*

2 Discrete distributions: how the probability is shared out

For a discrete X , its **probability mass function** (PMF) lists how much probability sits on each value.

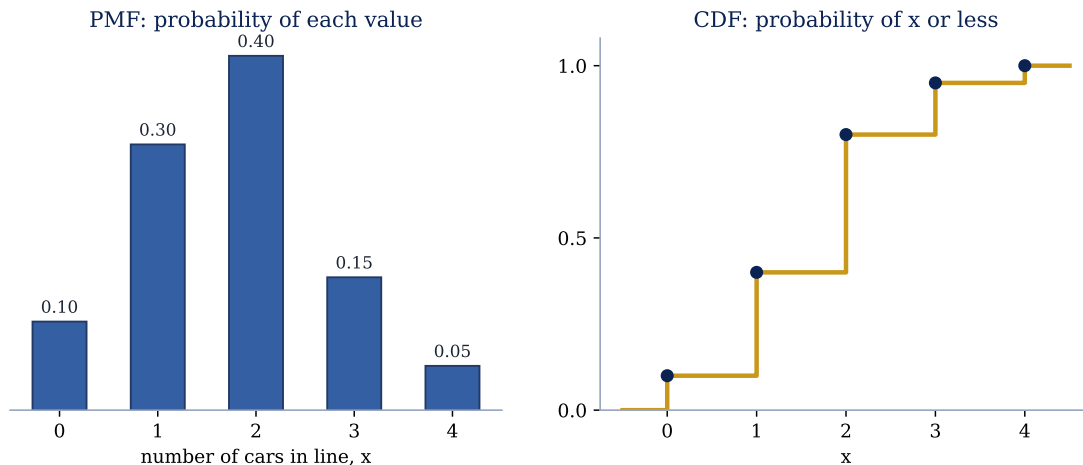
Key formula 2.1: Rules every PMF obeys

A function $f(x) = P(X = x)$ is a valid PMF when

$$f(x) \geq 0 \quad \text{for all } x, \quad \sum_x f(x) = 1.$$

Probabilities are never negative, and the total probability across all values is exactly one.

The simplest case is the **discrete uniform** distribution, where every value is equally likely: $f(x) = \frac{1}{n}$. A fair die is the classic example, $f(x) = \frac{1}{6}$ for $x = 1, \dots, 6$.



Left: a PMF places a bar of probability on each value (here, the number of cars at a drive-thru). Right: the cumulative distribution function (CDF) adds those bars up from the left, climbing in steps to 1 — it answers “what is the chance of x or fewer?”

Definition 2.1: Cumulative distribution function (CDF)

The **CDF** accumulates probability up to a value:

$$F(x) = P(X \leq x) = \sum_{x_i \leq x} f(x_i).$$

It starts at 0, only ever rises, and ends at 1 (because all the probability has been collected).

Worked example 2.1: Building a CDF from a PMF (cars in line)

With $P(0) = .10$, $P(1) = .30$, $P(2) = .40$, $P(3) = .15$, $P(4) = .05$, accumulate from the left:

$$F(0) = .10, \quad F(1) = .40, \quad F(2) = .80, \quad F(3) = .95, \quad F(4) = 1.00.$$

So $P(2 \text{ or fewer cars}) = F(2) = 0.80$. The CDF answers “up to here” questions directly, which is why software reports it constantly — percentiles, p -values and quantiles are all just the CDF read backwards.

3 Expected value: the balance point of a distribution

The **expected value** (or mean) of a random variable is its long-run average — the probability-weighted centre of all its values.

Key formula 3.1: Expectation, variance, standard deviation

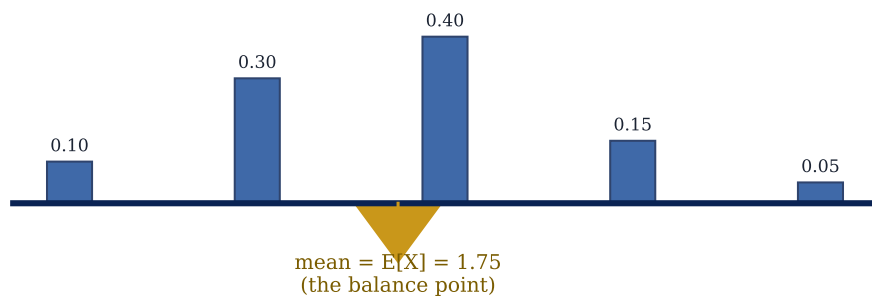
$$E[X] = \mu = \sum_x x f(x), \quad \text{Var}(X) = \sigma^2 = E[(X - \mu)^2] = \sum_x (x - \mu)^2 f(x) = E[X^2] - \mu^2,$$

$$\text{and } SD(X) = \sigma = \sqrt{\text{Var}(X)}.$$

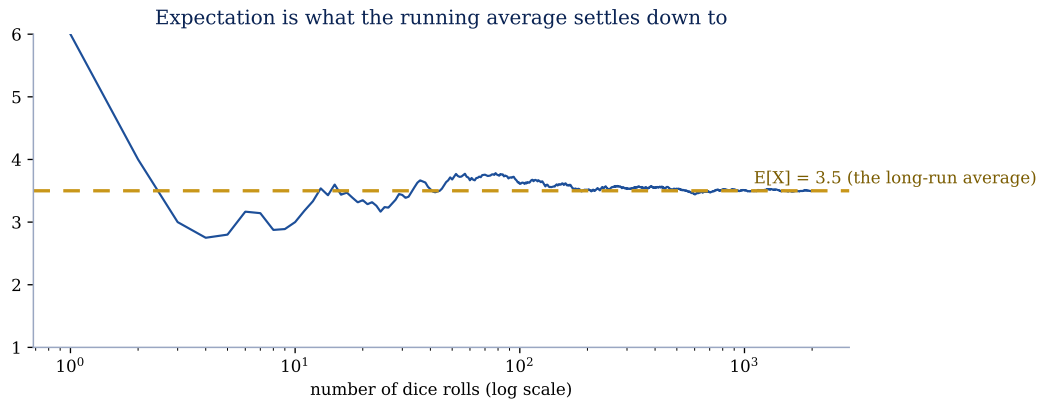
What each one is telling you

Expectation $E[X]$ is the *long-run average*: repeat the experiment thousands of times, average the results, and you land near $E[X]$. It need not be a value X can actually take (a family can have $E[X] = 2.3$ children). **Variance / SD** answers “how surprised should I be by a single outcome?” A small SD means outcomes huddle near the mean (predictable); a large SD means they scatter (risky). Notice the deep echo of Session 1: this is the *same* mean-and-spread story, now weighted by *probability* instead of by *counts*.

The expected value is where the distribution balances



The expected value is literally the balance point of the distribution. Stack each probability as a weight at its value; the mean is the fulcrum where the whole arrangement balances. This is the same “balance point” idea as the mean in Session 1, now weighted by probability instead of by counts.



Why we call $E[X]$ the “long-run average.” Rolling a fair die again and again, the running average wanders at first but steadily settles onto $3.5 = E[X]$. A single roll is unpredictable; the average of many is almost certain. (This is the famous Law of Large Numbers in action.)

Is the game fair?

A game is **fair** when your expected gain is exactly zero — on average you neither win nor lose. Expectation is how a casino (or an insurer) prices risk: they make sure *their* expected gain is positive, even though any single round is uncertain.

Worked example 3.1: A fair dice game

A rolls two dice. If the sum is 7, A wins \$3; if 8, loses \$2; if 3, loses \$4; otherwise nothing changes hands. Let X be A's gain. Using the sum probabilities:

$$E[X] = (+3)\frac{6}{36} + (-2)\frac{5}{36} + (-4)\frac{2}{36} = \frac{18 - 10 - 8}{36} = \frac{0}{36} = 0.$$

The expected gain is zero, so the game is **fair** — neither player is favoured in the long run.

Worked example 3.2: Mean and spread of cars in line

Cars in line at lunch follow $P(0) = .10$, $P(1) = .30$, $P(2) = .40$, $P(3) = .15$, $P(4) = .05$. The mean is

$$E[X] = 0(.10) + 1(.30) + 2(.40) + 3(.15) + 4(.05) = 1.75 \text{ cars.}$$

For the spread, $E[X^2] = 0 + 1(.30) + 4(.40) + 9(.15) + 16(.05) = 4.05$, so $\text{Var}(X) = 4.05 - 1.75^2 = 0.9875$ and $\text{SD}(X) \approx 0.99$. On a typical lunch you'll see about 1.75 ± 1 cars.

4 Continuous variables: probability becomes area

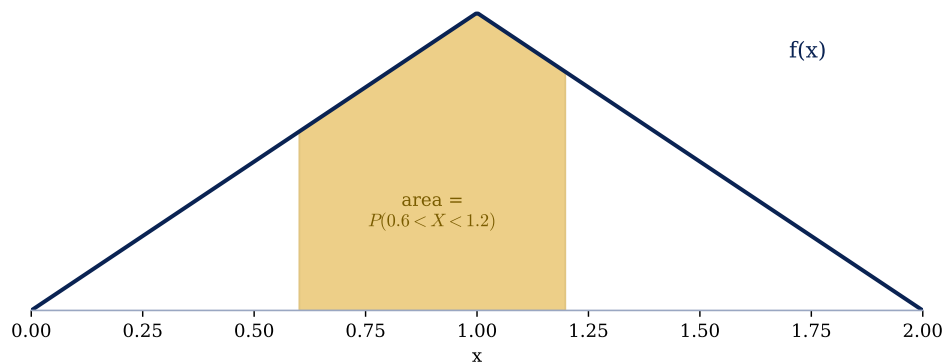
For a continuous X , asking “what is $P(X = 2.000\dots)$?” is meaningless — there are infinitely many possible values, so any single one has probability zero. Instead we ask about *intervals*, and probability becomes **area under a curve**.

Definition 4.1: Probability density function (PDF)

A continuous X is described by a density $f(x) \geq 0$ with total area 1. The probability of landing in an interval is the area beneath the curve there:

$$P(x_1 < X < x_2) = \int_{x_1}^{x_2} f(x) dx.$$

For continuous variables, probability = area under the curve



For a continuous variable, the density curve itself is not a probability — the shaded area between two points is. The whole area under the curve is always 1.

Mass along a rod

Picture probability as modelling clay smeared along a rod. For a discrete variable the clay sits in neat lumps at certain points (the PMF). For a continuous variable it is spread smoothly (the PDF), and the “amount” between two marks is the clay *in that stretch* — an area, never the height at a single point.

Worked example 4.1: A triangular density, fully worked

Let $f(x) = x$ for $0 < x < 1$ and $f(x) = 2 - x$ for $1 < x < 2$ (a tent shape). **Is it valid?** The total area is a triangle of base 2 and height 1, namely $\frac{1}{2}(2)(1) = 1$ — yes. **Find** $P(0.6 < X < 1.2)$ as the area in that strip, splitting at the peak $x = 1$:

$$\int_{0.6}^1 x \, dx + \int_1^{1.2} (2 - x) \, dx = \left[\frac{x^2}{2} \right]_{0.6}^1 + \left[2x - \frac{x^2}{2} \right]_1^{1.2} = 0.32 + 0.18 = 0.50.$$

By the tent's left-right symmetry the balance point is the middle, so $E[X] = 1$. Notice we *integrated* where, for a discrete variable, we would have *summed* — the same idea in its smooth form.

5 Two variables at once: joint, marginal, conditional

Often two random quantities travel together — a person's education *and* age, defects from two machines. Their **joint distribution** $f(x, y) = P(X = x, Y = y)$ gives the probability of each pair.

				row sums = marginal of X	
Education X	None	0.01	0.02	0.05	0.08
	Primary	0.03	0.06	0.10	0.19
	Secondary	0.18	0.21	0.15	0.54
	College	0.07	0.08	0.04	0.19
		0.29	0.37	0.34	1.00
	30	45	70		
	Age Y				

A joint distribution table: each inner cell is the probability of a specific (education, age) pair. Summing across a row or down a column collapses one variable away, leaving the marginal distribution of the other (shown in teal in the margins) — which is why it earns that name.

Definition 5.1: Marginal, conditional, independence

From a joint distribution:

- the **marginal** of X sums out Y: $f_X(x) = \sum_y f(x, y)$ (a row/column total — “X on its own”);
- the **conditional** of Y given $X = x$ is $f_{Y|X}(y) = \frac{f(x, y)}{f_X(x)}$ (the Session 3 idea, applied to variables);
- X and Y are **independent** if the joint factors into the marginals, $f(x, y) = f_X(x) f_Y(y)$ for all x, y .

Why “marginal”?

The name is delightfully literal: these totals are written in the *margins* of the joint table. Reading a margin throws away all information about the other variable and tells you about one alone — exactly what you’d want if you only cared about education, ignoring age.

Worked example 5.1: Flu given fever

Past data gives a joint distribution of Disease $\in \{\text{Flu}, \text{No Flu}\}$ and Symptom $\in \{\text{Fever}, \text{No Fever}\}$. To find $P(\text{Flu} | \text{Fever})$, take the joint probability of (Flu and Fever) and divide by the *marginal* probability of Fever:

$$P(\text{Flu} | \text{Fever}) = \frac{P(\text{Flu}, \text{Fever})}{P(\text{Fever})} = \frac{P(\text{Flu}, \text{Fever})}{P(\text{Flu}, \text{Fever}) + P(\text{No Flu}, \text{Fever})}.$$

The denominator is just the “Fever” column total — conditioning is, once again, picking the right denominator.

Worked example 5.2: Marginals, a conditional, and an independence check (education vs age)

Using the joint table in the figure above: the **marginal** of “College” is its row total, $f_X(\text{College}) = 0.07 + 0.08 + 0.04 = 0.19$. The **conditional** chance of age 30 given a college graduate is

$$P(\text{age 30} | \text{College}) = \frac{0.07}{0.19} \approx 0.37.$$

Independent? The age-30 marginal is $f_Y(30) = 0.29$, so the “if independent” prediction would be $f_X(\text{College}) f_Y(30) = 0.19 \times 0.29 \approx 0.055$. But the true joint value is $0.07 \neq 0.055$, so education and age are **dependent** — knowing someone’s education shifts the odds of their age bracket.

Your turn

1. A student passes a driving test in 1, 2, or 3 attempts with probabilities proportional to a given rule. Check that the probabilities sum to 1, then find $E[X]$ (the expected number of attempts).
2. A continuous density is $f(x) = x$ for $0 < x < 1$ and $f(x) = 2 - x$ for $1 < x < 2$ (a triangle). Verify the total area is 1, then find $P(0.6 < X < 1.2)$ as an area. What is $E[X]$ by symmetry?
3. For a joint table of X (defects from Machine A) and Y (defects from Machine B), compute both marginals, check whether X and Y are independent, and find $E[XY]$.

Where this leads

You now have the full descriptive–probabilistic toolkit: data summaries (Session 1), the rules of chance (Session 2), updating with information (Sessions 3–4), and now random variables and their distributions (Session 5). These distributions — uniform, and soon the binomial, Poisson and the famous Gaussian — are the vocabulary for the rest of the course: **hypothesis testing**, **prediction**, and **forecasting**. The story of describing data has become the story of modelling it.